AP CALCULUS AB • LESSON WARMUP

Derivative Rules Review

NOV 2, 2026

Name: _____ **Period:** _____

Warmup 1

No Calculator

3.7 Implicit differentiation

For the curve $x^2 + xy + y^2 = 12$, what is $\frac{dy}{dx}$ at $(2, 2)$?

A -2
B -1
C $-\frac{1}{2}$
D 1
E 2

Warmup 2

Calculator Permitted

3.5 Trig + product rule

Let $f(x) = x \sin x$. What is $f'(0.8)$ to the nearest thousandth? Use radians.

A 0.557
B 0.717
C 1.275
D 1.574
E 2.075

Product Rule

Differentiate each product. Show the two product-rule terms before simplifying.

1 Find $f'(x)$ for $f(x) = (x^2 + 3)(4x - 1)$.

2 Find $g'(x)$ for $g(x) = x^3 \cos x$.

3 If $h(x) = (x^2 - 1)\sin x$, find $h'(0)$.

4 The function $p(x) = (3x^2 + 2)(x^3 - x)$. Find the slope of its tangent line at $x = 1$.

Quotient Rule

Differentiate the numerator and denominator separately; keep the denominator squared.

5 Find $r'(x)$ for $r(x) = \dfrac{2x^2 + 1}{x - 3}$.

6 Find $q'(x)$ for $q(x) = \dfrac{\sin x}{x^2 + 1}$.

7 If $m(x) = \dfrac{x^3 - 2x}{x^2 + 2}$, find $m'(0)$.

8 For $k(x) = \dfrac{x + 1}{x^2 - 4}$, find the slope of the tangent at $x = 3$.

Chain Rule

Identify the inside function, then multiply by its derivative.

9 Find $u'(x)$ for $u(x) = (5x^2 - 2x + 1)^4$.

10 Find $v'(x)$ for $v(x) = \sin(3x^2 - x)$.

11 Find $w'(x)$ for $w(x) = \sqrt{2x^3 + 5}$.

12 Find $j'(x)$ for $j(x) = \left(1 + \cos(x^2)\right)^3$.

Implicit Differentiation

Treat y as a function of x. Include $\frac{dy}{dx}$ when differentiating terms in y.

13 For $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

14 For $x^3 + y^3 = 9xy$, find $\frac{dy}{dx}$.

15 The curve $x^2 + xy + 2y^2 = 16$ passes through $(2, 2)$. Find its tangent slope there.

16 For $\sin(xy) + y^2 = x$, find $\frac{dy}{dx}$.